

Unneeded faces inside comparator models

[MC-177453](#) TEXTURES AND MODELS

Status	Resolved / Fixed
Affected	1.15.2; 20w14a; 20w18a; 20w21a; 1.16 Pre-release 2; 1.16 Pre-release 3; 1.16.1; 20w27a; 20w28a; 20w29a; 20w30a; 1.16.2 Pre-release 1; 20w49a; 20w51a; 21w03a; 1.16.5; 21w05b; 21w06a; 21w08b; 21w15a; 21w16a; 21w20a; 1.17 Pre-release 1; 1.17; 1.17.1; 1.18; 1.18.1; 1.18.2; 1.19; 1.19.2 (+more)
Fixed in	24w33a
Reported	2020-04-04
Mojira category	Textures and models

REPRODUCTION TARGET

Version: Minecraft Java Edition 1.21 (an affected release; do not use snapshot 24w33a or a later version for this reproduction)

Environment: Vanilla Minecraft Java Edition 1.21 in a single-player Creative-mode world with commands enabled. Use the default resource pack and no mods, shaders, or resource packs that replace the redstone comparator block model.

PRECONDITIONS

- Launch the game using the exact Java Edition 1.21 installation selected in the launcher.
- Create or load a single-player world in Creative mode with commands enabled. No particular world seed, coordinates, biome, or difficulty is required by the report.
- Keep the default vanilla resource pack active. In particular, do not install the corrective resource pack referenced by the report or any other pack that replaces comparator model files.
- Use an open area containing a flat, solid upper block face on which a comparator can be placed.
- Keep powered redstone components, redstone blocks, levers, redstone torches, and other possible signal sources away from the comparator so it remains unpowered.
- The input names below assume default controls. If controls have been rebound, use the bound inputs for Inventory, Open Chat, movement, jump/ascend, and sneak/descend.

STEPS TO REPRODUCE

1 **REPORT**

Start Minecraft Java Edition using the 1.21 installation, open the prepared Creative-mode single-player world, and wait for world loading to finish.

SUCCESS CUE The player can move normally, nearby terrain has rendered, and no loading screen is present.

2 **INFERRED**

Open the Creative inventory using the Inventory control.

INPUT E

SUCCESS CUE The Creative inventory interface is open and item category tabs are visible.

3 **INFERRED**

Open the Search Items tab, select its search field, and search for the redstone comparator item.

INPUT

Click the Search Items tab (compass icon), click the search field, and type: redstone comparator

SUCCESS CUE A Redstone Comparator item appears in the filtered Creative inventory results.

4 **INFERRED**

Move one Redstone Comparator from the search results into an empty hotbar slot. The hotbar is the row of nine slots along the bottom of the inventory interface.

INPUT Left-click the Redstone Comparator result, then left-click an empty hotbar slot

SUCCESS CUE A Redstone Comparator icon is present in one of the nine hotbar slots.

5 **INFERRED**

Close the inventory and select the hotbar slot containing the Redstone Comparator.

INPUT

Press E, then press the corresponding number key 1 through 9 or use the mouse wheel until the comparator slot is highlighted

SUCCESS CUE The inventory is closed, the comparator's hotbar slot is highlighted, and the comparator is shown as the selected item.

6 **INFERRED**

Move to an isolated flat, solid block face. Confirm that there are no powered redstone components immediately behind, beside, or underneath the intended comparator position.

INPUT Use W/A/S/D and the mouse to move and look

SUCCESS CUE The crosshair can be aimed at the top of a solid block with no nearby source capable of supplying redstone power to the comparator.

7 **REPORT**

Aim at the top face of the solid block and use the selected comparator to place it.

INPUT Right-click the top face of the solid block

SUCCESS CUE A redstone comparator is placed upright on the block and its three small torch positions are visible on top.

8 **REPORT**

Leave the newly placed comparator untouched in its default comparison mode and confirm that it is not receiving a redstone signal. Do not right-click it, because that would change its mode.

SUCCESS CUE The comparator remains isolated, its output is inactive, and its torches are not displaying a powered bright-red state.

9 **INFERRED**

Enter Spectator mode by executing the game-mode command in in-game chat. Include the leading slash because the command is being entered by a player rather than through a dedicated-server console.

INPUT Press T, type `/gamemode spectator`, then press Enter

SUCCESS CUE A game-mode confirmation appears and the player is in Spectator mode, allowing free flight and movement through blocks.

10 **REPORT**

Fly next to the comparator and lower the camera to the comparator model's height. Face the comparator from one side and move into it using short movement taps so the camera clips through the model rather than passing completely beyond it.

INPUT

Use the mouse to look; tap W/A/S/D to move; hold or tap Left Shift to descend; use Space to ascend if the camera goes too low

SUCCESS CUE The Spectator camera is intersecting the comparator's low-profile model and can move through it without collision.

11 **REPORT** **TRIGGER**

Fine-adjust the clipped camera position until it is inside the comparator body and directly below one or more of the unlit torch cubes. Angle the camera upward toward the underside of those torch cubes. Stop moving as soon as this viewing position is established.

INPUT

Use short taps of W/A/S/D, Space, and Left Shift while moving the mouse upward toward the unlit torch geometry

SUCCESS CUE The camera is inside the comparator model, one or more unlit torch cubes are directly above the viewpoint, and the view is aimed at their bottom surfaces.

HANDOFF TO OBSERVATION

The Spectator camera has just been positioned inside the unpowered comparator, directly below one or more unlit torch cubes, with the view aimed upward at their undersides. Keep the camera stationary and hand control to the observation subsystem.

UNCERTAINTIES

- The report supplies no seed, coordinates, or required comparator orientation; these do not appear necessary because the model can be inspected in any world and orientation.
- The exact camera offset is not specified. Small movement and viewing-angle adjustments may be required because clipping and camera sensitivity vary, but the final position must remain inside the comparator and below the unlit torch geometry.
- The corrective resource pack referenced through MC-214686 is not part of the reproduction setup and must not be enabled when reproducing the vanilla 1.21 model.

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Trigger reached?	YES / NO
Crash file	
Notes	
Tested by / date	